The Twin Criterion in Cell Coordinates

IV. What the framework proves about twin pairs

Abstract

This paper collects what the cell coordinates prove outright about the twin problem. Everything below is proved, with one exception flagged as such (Verified Law 17, proved under a hypothesis on prime gaps and verified beyond it); the measurements, the test cases and the account of where the framework stops are Paper V.

The gap between consecutive odd composites takes only the values 2, 4 and 6, and gap 6 *is* a twin pair (Theorem 1), so the twin conjecture becomes the statement that the maximal possible gap is attained infinitely often — a cap that is free and proved. The ladder below it is complete: gap 2 occurs infinitely often with no prime input, gap 4 with one use of Dirichlet, and gap 6 needs two primes simultaneously (Theorem 2). No five survivors lie in arithmetic progression at spacing 6 on one rail, beyond a single small exception, and the full-cycle run counts are $\prod(q-k)$ (Theorem 3, Corollary 1).

Inside a single sector the criterion sharpens as far as the framework can take it: after switching on every line $p \leq M$, **at most six cells can be open without being a twin pair, and their positions are given by an explicit formula in** M (Theorem 4). The lines 5 and 7 alone cut six to three, and no further small line lowers that; but the three occur only when $(M+2, M+4)$ is itself a twin, so a twin follows from three open cells rather than seven (Corollary 2). Taken over a whole period of 35 sectors the budget falls to 31, and §3.3 shows that no finite set of lines lowers it further: one configuration survives every residue argument, and it is admissible at every prime. A twin therefore follows from one open cell that avoids six named places — the places being fixed by the geometry of the square rather than chosen by the lines.

The closing budget is exact: the distance-6 graph has minimum deletion cover $\tau = T - U + Q$ (Theorem 8), the four-state census transports under a new line (Theorem 9), and the late lines of a sweep compress (Theorem 10). The clock coordinate turns primality into a zero-test (Theorem 11) and forces every surviving cofactor in a range to be prime (Theorem 12). Read along a whole row that becomes a general law: for $p \leq p + 2j < p^2$ the cell $p(p+2j)$ is **original** — owned by no smaller line — exactly when $p+2j$ is prime (Theorem 13), so a row carries the primality of every odd number up to p^2 in one bit per cell, the twin condition becomes "originality survives one step past the diagonal", and the cell it tests is $(p+1)^2 - 1$, which is [I, Thm 5] read along the row. Finally, across the *sequence* of sectors, a fixed set of lines leaves exactly $12S$ fresh open cells each time its cycle returns to the same phase (Theorem 14); a later line closes at most two of the twelve copies of any surviving class and at most one beyond an explicit threshold (Theorem 15, Corollary 7).

A last section takes the four cells nearest the window's ends and turns them into four **tracks** as a runs, each member a quadratic in a; a prime can own a track only if the corresponding discriminant is a quadratic residue (Theorem 18), which makes the four tracks unequal — Bateman–Horn gives them densities 0.41, 0.73, 1.10 and 1.43 times that of a generic cell, so the last is three and a half times richer in twins than the first. And where eligibility admits three quarters of the primes for one track and a quarter for all four, **simultaneous double duty inside one window admits exactly nine primes, so every** $r > 97$ **can close at most one of the four** (Theorem 19).

That constraint is sharp for a single line and empty for four: assigning a distinct line to each track and combining the congruences produces an explicit infinite progression along which all four cells are closed, q is prime and $q+2$ is composite (Proposition 3), so **no fixed number of named cells can force a twin.** The same coordinates make the operative threshold visible: a line whose modulus reaches the window's own length cannot wrap inside it and has a total budget of two, so at $a = 1667$ the 857 lines below N close 6,500 cells while the 370 above it close 20.

A further section changes the unit from the sector to the **belt** between consecutive prime gates q^2 and r^2. The belt holds exactly $(r^2 - q^2)/6 - 1$ cells and so grows like qg with $g = r - q$ (Theorem 16), while the line born at its left end can strike only $\lfloor 2g/3 \rfloor$ of them — **a count independent of** q (Verified Law 17, proved under $g^2 < 2q$). Measured, the new line closes nothing at all in about three quarters of belts. A deterministic ceiling for a whole age layer follows (Proposition 2), generous by one to two orders of magnitude; but summed over the full pyramid of layers it exceeds the belt by a factor $\sim \frac{2}{3}\log q$, so the recent lines are not capacity-limited and the argument does not close. We report that negative outcome with the rest.

No progress toward the twin-prime conjecture is claimed. Every statement here is exact; none of them is a lower bound on anything.

Keywords: twin primes, gap alphabet, minimum vertex cover, sieve cycles, cell coordinates.

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1. Setting

We use Papers I–III as follows and import nothing else.

- **[0]** supplies the window combinatorics: the increments W_j of $\lfloor 2j^2/p \rfloor$, their uniform bound and their exact histogram.
- **[I]** supplies the coordinates: the line $L_p(k) = p(p+2k)$ beginning at p^2, the grid L_3, the cells $C_b = (6b-1, 6b+1)$, and the square window with its $+8$ growth.
- **[II]** supplies the exact cycle laws: the four-state law, its refinement by inheritance depth, arbitrary weights in $\Omega_{\le z}$, the refinement by line size, and the disjoint ownership layers.
- **[III]** studies the transfer to a window of length $\asymp z^2$. On the tested windows, linear depth weights measure 1.0000 to the reported precision, the tested soft truncations remain close to 1, and the sharp depth-zero indicator is the exceptional endpoint with ratio near 0.80. These are measurements, not an exact window theorem.

A word on the last point, since it is what makes this paper's organisation possible. The indicator of depth zero is the condition "both members of the cell survive" — that is, the twin condition. Thus, in the tested family, the largest transfer loss occurs precisely at the depth-zero indicator, the quantity that becomes a primality/twin condition inside the moving square window.

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2. The gap alphabet and the ladder of proofs

Before turning to the deviation analysis we record what the framework proves outright about gaps, and where the ladder stops.

2.1 Theorem 1 (the gap alphabet)

Theorem 1. The gap between consecutive odd composites takes only the values 2, 4 or 6.

Proof. Among any three consecutive odd numbers $n, n+2, n+4$ the residues mod 3 are a permutation of $\{0, 1, 2\}$, so one is divisible by 3; and every odd multiple of 3 from 9 onward lies on L_3. Hence beyond 7 no three consecutive survivors exist, and the gap is capped at 6. ■

Correspondingly: gap 2 means no survivor between; gap 4 means one isolated prime; and **gap 6 means two adjacent survivors, i.e. a twin pair.** Verified: every gap-6 interval contains a twin, zero failures among 2,992 instances below 3×10^5.

Measured over 9×10^8 gaps up to 2×10^9, the three letters occur with frequencies 0.89816, 0.09475 and 0.00708; the maximum observed gap is 6, first attained at 9.

Hence an equivalent form of the twin conjecture: **the maximal possible gap is attained infinitely often.** The cap is free and proved; the statement concerns composites, which are the objects the framework actually constructs.

2.2 The ladder, and where it stops

Theorem 2. Gap 2 occurs infinitely often, requiring no prime; gap 4 occurs infinitely often, requiring one prime.

Proof. For gap 2: take $n = 30j + 3$ and $n + 2 = 30j + 5$; the first lies on L_3, the second on L_5, both for every $j \geq 1$. For gap 4: let $p \equiv 8 \pmod{15}$ be prime; then $p - 2 \equiv 6$ is divisible by 3 and $p + 2 \equiv 10$ by 5, so p is an isolated survivor, and Dirichlet's theorem supplies infinitely many such p. ■

gap 2	zero primes required	**proved**
gap 4	one prime (Dirichlet)	**proved**
gap 6	**two primes simultaneously**	**open**

Dirichlet supplies one prime in an arithmetic progression; nothing supplies two at a prescribed distance. That is the whole of the remaining distance.

Why this reciprocal-sum test cannot bridge it. Summing reciprocals by gap type:

N	gap 2	gap 4	gap 6
10^6	2.805	0.887	0.464
10^8	4.556	1.127	0.488
2×10^9	5.762	1.257	**0.498**

Gaps 2 and 4 have **divergent** reciprocal sums (growing like $\log\log N$), and divergence proves infinitude. The gap-6 sum **converges** — flattening toward a constant. That constant is the analogue of Brun's constant in the present alphabet and not Brun's constant itself: the latter is $\sum_{\text{twins}} (1/p + 1/(p+2)) = 1.9021605\ldots$, whereas the column above carries one reciprocal per gap-6 event. A convergent reciprocal sum cannot distinguish "infinitely many" from "finitely many". Thus this particular divergence-based density test has no route to the twin conclusion.

2.3 Runs and the cap at four

> **Theorem 3.** On a single rail, no five survivors with $x > 5$ can lie in arithmetic progression with common difference 6; and $5, 11, 17, 23, 29$ is the only exception.

Proof. Since $6 \equiv 1 \pmod 5$, the five terms $x, x+6, \ldots, x+24$ run through all residues modulo 5, so exactly one of them is divisible by 5. If that term exceeds 5 it is an odd multiple of 5 at least 25, hence lies on L_5 and is struck. The term can fail to be struck only when it equals 5 itself, which forces $x = 5$. ■

The exception is real and must be carried in the statement. For $x = 5$ the run is $5, 11, 17, 23, 29$, and all five are prime; an exhaustive search to 2×10^5 finds this and no other. It exists purely because L_5 is born at 25: as a statement about the *residue classes* the argument is exact, and only the birth rule creates the exception.

Verification. At sieve depth 97 over 6×10^7 cells, the run-length census is 6,448,150 of length 1; 2,404,092 of length 2; 778,762 of length 3; 211,170 of length 4; and **none of length** 5 **or more.**

> **Corollary 1.** With G_k the number of runs of k consecutive survivors at spacing 6 on one rail, the full-cycle counts satisfy $G_k = \prod_q (q-k)$.

Proof. A run of k forbids exactly k marks on each ruler, leaving $q - k$. ■

Verification (direct count on the t-axis at $p = 101$):

lines	k=1	2	3	4	**5**
$\{5\}$	4	3	2	1	**0**
$\{5, 7\}$	24	15	8	3	**0**
$\{5, 7, 11\}$	240	135	64	21	**0**

The four laws $(q-1), \ldots, (q-4)$ are therefore not four phenomena but one ruler with different numbers of marks.

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3. The twin problem inside the framework

3.1 A surviving cell is a twin pair

Let $u < v$ be consecutive integers coprime to 6. Every survivor of all lines $\le u$ inside (u^2, v^2) is necessarily prime, since a composite $x < v^2$ has a prime factor $\le \sqrt{x} < v$, and there is no prime strictly between u and v. Hence

$$\boxed{\text{a surviving } NN \text{ cell in } (u^2, v^2) = \text{a twin prime pair.}} \qquad (3.1)$$

3.2 Theorem 4: only six cells in a sector can be open without being a twin

Take the sector in the form used throughout this section: $M \equiv 3 \pmod 6$ and the interval $(M^2, (M+6)^2)$, which carries $A_M = 2M + 6$ cells

$$C_j = \left(M^2 + 6j + 2,\ M^2 + 6j + 4\right), \qquad j = 0, 1, \ldots, 2M + 5.$$

Theorem 4. Switch on every line $p \le M$. Then at most **six** cells of the sector can be open without being a twin pair, and their positions are given explicitly by

$$E_M = \underbrace{\left\{\tfrac{2M}{3},\quad M+1,\quad \tfrac{5M}{3}+2,\quad 2M+3\right\}}_{\text{present only if } M+2 \text{ is prime}} \cup \underbrace{\left\{\tfrac{4M}{3}+2,\quad 2M+5\right\}}_{\text{present only if } M+4 \text{ is prime}}.$$

Consequently **every open cell whose index lies outside E_M is a twin pair.**

Proof. Let N be a composite endpoint of an open cell. Every prime factor of N exceeds M, since otherwise some line $p \le M$ would have closed it; and the least prime factor cannot be $\ge M+6$, since then $N \ge (M+6)^2$. As $M \equiv 3 \pmod 6$ and N is coprime to 6, the least prime factor is therefore $M+2$ or $M+4$.

For $p = M+2$ the strikes below $(M+6)^2 = (p+4)^2$ are $p^2,\ p(p+2),\ p(p+4),\ p(p+6),\ p(p+8)$ — the next one, $p(p+10)$, already exceeds $(p+4)^2$. Of these, $p(p+4) = (M+2)(M+6)$ is divisible by 3 and so lies on L_3, not in a cell; four remain. For $q = M+4$ the strikes below $(q+2)^2$ are $q^2,\ q(q+2),\ q(q+4)$, of which $q(q+2) = (M+4)(M+6)$ lies on L_3; two remain. Six in total.

Their positions follow by writing each product in the form M^2+6j+2 or M^2+6j+4. For instance $(M+2)^2 = M^2+4M+4 = M^2+6j+4$ with $j = 2M/3$, and $(M+2)(M+8) = M^2+10M+16 = M^2+6j+4$ with $j = 5M/3+2$; each is an integer because $3 \mid M$. Finally, if $M+2$ is composite then it has a prime factor $\le M$, so its four products were already closed by an older line and are not exceptions; likewise for $M+4$. ■

Verification. Zero violations over every $M = 9, 15, \ldots, 19{,}999$ — 3,332 **sectors.** In each, every open cell outside E_M was checked to be a twin.

Example, $M = 9$**.** The sector $(81, 225)$ has 24 cells; $E_9 = \{6, 10, 14, 17, 21, 23\}$, carrying

$$121 = 11^2,\quad 143 = 11\cdot 13,\quad 169 = 13^2,\quad 187 = 11\cdot 17,\quad 209 = 11\cdot 19,\quad 221 = 13\cdot 17.$$

No other composite can inhabit an open cell of that sector.

The count of exceptions is 0**,** 2**,** 4 **or** 6, according to the primality of $M+2$ and $M+4$:

$M+2$	$M+4$	maximum exceptions
composite	composite	0
prime	composite	4
composite	prime	2
prime	prime	6

At $M = 999$ both $1001 = 7\cdot 11\cdot 13$ and $1003 = 17\cdot 59$ are composite, so E_M is empty and **all** 93 **open cells of that sector are twin pairs** — as measured.

What this buys, stated precisely. Writing C_M for the number of open cells, one has $T_M \ge C_M - |E_M| \ge C_M - 6$, so a twin follows from $C_M \ge 7$. Theorem 4 replaces that by the much weaker requirement

$$S_M \not\subseteq E_M, \qquad (3.2)$$

where S_M is the set of open indices: **a single open cell suffices, provided it is not at one of six named places.** And the six places are fixed by the geometry of the square — they are not chosen by the lines, whose phases are periodic and unrelated to M.

The lines 5 and 7 alone cut six to three, and the pair is not interchangeable. Writing $M = 6n + 3$, the six positions are exact quadratics in n as absolute cell indices:

$$A = 6n^2{+}10n{+}4, \quad B = 6n^2{+}12n{+}6, \quad C = 6n^2{+}14n{+}8, \quad D = 6n^2{+}16n{+}9, \quad E = 6n^2{+}18n{+}11, \quad F = 6n^2{+}18$$

(verified for $n = 1, \dots, 3999$ against the definition). Each is therefore a function of n mod p modulo any p, so the state of all six under the lines 5 and 7 depends on n mod 35 alone: **the 35 classes are not a sample but the whole question**, and checking them is a proof rather than an experiment. Doing so gives $\max |S_M \cap E_M| \le 3$ after those two lines alone, with only four maximal patterns —

$$\{A, B, C\}, \quad \{A, E, F\}, \quad \{C, D, F\}, \quad \{C, E, F\},$$

writing $A, \dots, F$ for the six positions in the order listed above. The collapse from $2^6 = 64$ conceivable patterns to four is what makes the next step possible.

The pair is special. Line 5 alone leaves four open; adding 7 gives three; and **adding 11, then 13, changes nothing at all** — the same bound and the same four patterns. The reason is structural rather than numerical: a larger line can choose a phase striking none of the six, and the Chinese remainder theorem then combines that phase freely with the phase of 5 and 7 that leaves three. **So the route "add more small lines until the six are closed" is shut**, and we record it so that it is not attempted again in another notation.

> **Corollary 2 (the dichotomy).** Suppose $(M + 2, M + 4)$ is not a twin pair. Then $|S_M \cap E_M| \le 2$, and consequently
>
> $$C_M \ge 3 \quad \Longrightarrow \quad \text{the sector contains a twin, or } (M + 2, M + 4) \text{ is one.}$$

Proof. By Theorem 4 each of A, D, E is present only if $M + 2$ is prime and each of C, F only if $M + 4$ is prime, while B carries $(M + 2)(M + 4)$ and so requires both. If the open set met both groups, $M + 2$ and $M + 4$ would both be prime and $(M + 2, M + 4)$ would be the twin. So under the hypothesis the open set lies inside $\{A, D, E\}$ or inside $\{C, F\}$, and B is closed. Each of the four maximal patterns above meets both groups, hence none survives; inspecting the 35 classes, the patterns that do survive are $\varnothing$, $\{A\}$, $\{C\}$, $\{F\}$, $\{A, D\}$, $\{A, E\}$ and $\{C, F\}$, all of size at most two. ■

Verification. Over the same 3,332 sectors $M = 9, 15, \dots, 19{,}999$: the maximum of $|S_M \cap E_M|$ is 3 in the 340 sectors where $(M + 2, M + 4)$ is a twin and **exactly 2 in the other** 2,992; zero violations of the forcing (an open member of $\{A, D, E\}$ with $M + 2$ composite, or of $\{C, F\}$ with $M + 4$ composite, or B open without the twin), and zero cases where the open set met both groups without the twin being present. **The separation is exact: three open positions occur only when the twin is already there.** Both bounds are attained — three in 3 sectors, two in 13 of the twinless ones — so neither is an artefact of a range too short to reach them.

Five residue classes in which the two lines close all six. Of the 35 classes, five leave nothing open: $n \equiv 0, 8, 23, 25, 33 \pmod{35}$, that is

$$M \equiv 3,\ 51,\ 141,\ 153,\ 201 \pmod{210}.$$

There $S_M \cap E_M = \varnothing$ from the lines 5 and 7 alone, so **every** open cell is a twin. Condition (3.2) remains the weakest form of the criterion — it asks for no count at all — but in these classes it becomes *equivalent* to the bare $C_M \ge 1$, and Corollary 2 lowers the sufficient count in a general sector from $C_M \geq 7$ to $C_M \geq 3$. The improvement is to the numerical threshold, not to the criterion. Measured over the 713 such sectors below $M = 30{,}000$: 444,958 open cells and **not one of them fails to be a twin**, against 155 non-twin survivors in the 428 ordinary sectors below $M = 3000$ alone. And the restriction costs nothing in the count itself — the survivor density in these five classes, against all other classes, is 0.934 up to $M = 4000$, 0.997 up to 12000 and 1.000 up to 30000; fixing the phases of 5 and 7 changes which cells die, not how many.

And what it does not buy. C_M exceeds the twin count of the sector by at most six. Any lower bound on C_M is therefore a lower bound on twins, and (3.2) is an exact reformulation rather than a route. We record it because it is the sharpest form the framework has produced of the twin criterion, not because it weakens the problem.

3.3 The exception budget over a full period, and the limit of local arguments

Corollary 2 bounds the exceptions in one sector. Consecutive sectors are not independent, and taking a whole period of them together lowers the bound further — up to a point that can be identified exactly.

The sectors tile: $(M^2, (M+6)^2)$ is followed immediately by $((M+6)^2, (M+12)^2)$, so a prime forced near the top of one is forced near the bottom of the next. Taking the 35 sectors of one full period together — the window $(M_0^2, (M_0+210)^2)$ with $M_0 = 210k+3$ — and writing each exceptional position's requirement as a set of offsets that must be prime (A needs $M+2$; C needs $M+4$; D needs $M+2, M+8$; E needs $M+2, M+10$; F needs $M+4, M+8$), the twinless hypothesis forbids any two forced offsets differing by 2 across the whole window, not merely within a sector.

Summing the per-phase caps independently gives $5 \cdot 0 + 20 \cdot 1 + 10 \cdot 2 = 40$. The coupling costs three of those: three pairs of adjacent phases cannot both attain their caps, because the offsets clash across the join — for instance a phase reaching two via $\{C, F\}$ forces $M+8$, and its successor reaching two the same way forces $M'+4 = M+10$. **A fourth adjacent pair with the same caps does not clash**, which is the point: the obstruction is arithmetic in the offsets, not a consequence of adjacency. Hence 37, and with the lines $11, 13, 17$ admitted as well, an exhaustive scan over all $5 \cdot 7 \cdot 11 \cdot 13 \cdot 17/35 = 2431$ alignments gives

$$\sum_{i=0}^{34} |S_{M_i} \cap E_{M_i}| \leq 31 \qquad \text{(no twin in the window)},$$

the ceiling 31 being attained in 9 of the 2431 alignments, and the distribution of the 2431 values peaking at 26.

That ceiling cannot be lowered by any finite set of lines, and this is a theorem rather than a report of failed attempts. The 9 extremal alignments carry 28 distinct maximal configurations between them. Each configuration demands not only that certain $M_0 + a$ be prime — 27 to 30 of them — but also that the *partner* member of each of the 31 cells survive, and those partners are quadratics in n: $36n^2 + 60n + 23$ for A, $36n^2 + 84n + 47$ for C, $36n^2 + 96n + 53$ for D, $36n^2 + 108n + 67$ for E, $36n^2 + 108n + 79$ for F (verified against the definition for $n < 3000$). Testing the full linear-and-quadratic systems for a fixed prime divisor kills 27 **of the** 28: $q = 23$ disposes of fourteen, $q = 31$ of ten, $q = 19$ of three. One survives, at $M_0 \equiv 448353 \pmod{510510}$, with 28 linear and 31 quadratic conditions — 59 polynomials of total degree 90. Every prime $q \leq 90$

leaves at least one admissible residue (the tightest are $q = 19$ and $q = 31$, with exactly one each), and for $q > 90$ a residue survives automatically since 90 polynomials of that total degree have at most 90 roots. **The system is therefore admissible at every prime, so by the Chinese remainder theorem the residues can be chosen simultaneously against any finite list of further lines: no residue-based argument, however many primes it uses, reaches 30.**

Nor does the order in which the lines are born supply the missing constraint. One might hope that a line $M_0 + a$, forced prime by one exceptional cell, strikes another of the 31 once it is born. Modulo $p = M_0 + a$ one has $M_0 \equiv -a$, so a member $(M_0 + u)(M_0 + v) + e$ with $e \in \{0, \pm 2\}$ reduces to the integer $(u - a)(v - a) + e$, and $|u|, |v|, |a| < 216$ while $p \asymp M_0$; the line divides the member only if that small integer is exactly zero. Excluding the cell's own factors, this needs $|u - a| \cdot |v - a| = 2$, hence $|u - v| \in \{1, 3\}$. **But u and v are the offsets $6i + d$ with $d \in \{2, 4, 8, 10\}$, so $|u - v|$ is even, and the collision is impossible.** The check finds none, for the 28 lines or for any $M_0 + b$ with $b \le 216$. The timing of the lines carries no information the residues did not already carry, and the escapee's survival is equivalent to the simultaneous primality of its 59 irreducible polynomials — precisely the setting of Schinzel's Hypothesis H [V, ref. 27], which predicts infinitely many t realising it. **Proving the configuration impossible would mean contradicting that prediction for a specific family, which is not a weakening of the twin problem but an apparent strengthening of it.**

3.4 Blocks of consecutive periods, and why lengthening the block does not help

Section 3.3 bounds the exceptions over one period of 35 sectors. Consecutive periods are coupled in the same way that consecutive sectors are, so the same question can be asked of a block of L periods — $35L$ sectors — and the answer moves. Write B_L for the maximum of $\sum_i |S_{M_i} \cap E_{M_i}|$ over such a block under the twinless hypothesis, so that $B_1 = 31$ is the content of §3.3 and

$$\sum_i C_{M_i} > B_L \implies \text{a twin in the block.}$$

How B_L is computed, and a trap that is worth naming. The natural procedure — enumerate the maximal configurations of each residue class and test each for a fixed prime divisor — is *not* sound for this purpose. A residue class whose maximum is m also carries configurations of size $m - 1$, $m - 2$ and so on, obtained by deleting cells; deleting a cell removes conditions and can only make admissibility easier. Those sub-configurations are never enumerated, so "every configuration at level ℓ dies" establishes nothing about level ℓ. The sound procedure reverses the order: fix a residue for each prime q, switch the line q on *inside* the transfer recursion, and read off the maximum. That maximum is by construction the largest admissible configuration at those residues, and the maximisation over residues is B_L. It is also faster by orders of magnitude.

Values. By that procedure, maximising over the residues of every prime up to the stated bound:

$$B_1 = 31, \qquad B_3 = 67, \qquad B_5 = 100, \qquad B_7 = 138, \qquad B_9 = 163.$$

The first three are certain: B_1 is §3.3; B_3 and B_5 are stable under every prime tested and B_5 was obtained twice, once by the sound procedure and once by exhaustive enumeration with an independent admissibility test, which agree. B_7 and B_9 are stable across twelve and eight consecutive primes respectively but were computed with pruning, so they are lower bounds with strong stability rather than certified maxima.

The extremal configuration at $L = 3$ carries 66 linear and 67 quadratic conditions — 133 polynomials of total degree 200 — and every prime $q \le 200$ leaves it an admissible residue, so §3.3's conclusion transfers: no finite set of lines lowers 67 either. Two features of the optimum are worth recording. First, it does *not* preserve the single-window maximum in the middle: the three windows contribute $20, 26, 21$, so the block optimum spends $31 - 26 = 5$ of the central window's capacity to gain elsewhere. Second, the killing is concentrated: across every level from 144 down to 138 at $L = 7$, the primes 29 and 37 account for about ninety per cent of the eliminated configurations, the remainder falling to 43, 53, 71, 73, 79 and 97.

In requirement per sector, $(B_L + 1)/35L$ reads 0.914, 0.648, 0.577, 0.567, 0.521. It falls, and the next paragraph says why, and how far.

The quadratic shadow, and the order of B_L. Each exceptional cell carries, besides its linear primality conditions, the condition that the *other* member of the cell survive. Those partners are

$$Q_A = (M{+}2)^2{-}2, \quad Q_C = (M{+}4)^2{-}2, \quad Q_D = (M{+}5)^2{-}11, \quad Q_E = (M{+}6)^2{-}14, \quad Q_F = (M{+}6)^2{-}2,$$

so each type forbids, modulo a prime $q > 17$, a number of residue classes that is exactly

$$\omega_A = \omega_C = 2 + \chi_2(q), \qquad \omega_D = 3 + \chi_{11}(q), \qquad \omega_E = 3 + \chi_{14}(q), \qquad \omega_F = 3 + \chi_2(q),$$

with $\chi_d(q)$ the Legendre symbol. The linear conditions supply one forbidden class for A and C and two for D, E and F; the partner supplies two more exactly when the relevant d is a quadratic residue. Since the quadratic characters are non-principal, each ω has mean value 2 for A and C and 3 for D, E and F.

Proposition 1. Let a block of $N = 35L$ consecutive sectors carry an admissible configuration of exceptional cells under the twinless hypothesis, and for $X \in \{A, C, D, E, F\}$ let I_X be the set of sector indices carrying a cell of type X. Then

$$|I_A|, |I_C| \ll \frac{N}{\log^2 N}, \qquad |I_D|, |I_E|, |I_F| \ll \frac{N}{\log^3 N},$$

and consequently

$$B_L \ll \frac{L}{\log^2 L}.$$

Proof. **Step 1: each type forbids an exact number of residue classes.** Fix a prime $q > 17$. Admissibility at q means that some residue may be assigned to M_0 modulo q at which no polynomial of the configuration vanishes; fix that residue and write $t \equiv M_i \pmod q$. Since 6 is invertible modulo q, the map $i \mapsto t$ is a bijection of $\mathbb{Z}/q$, so it suffices to count forbidden values of t.

A cell of type A at index i carries the linear condition that $M_i + 2$ be prime and the quadratic condition that its partner $(M_i + 2)^2 - 2$ survive. Modulo q these forbid $t \equiv -2$, one class, and $(t + 2)^2 \equiv 2$, which has $1 + \chi_2(q)$ solutions with $\chi_d(q)$ the Legendre symbol. The two never coincide, since $t \equiv -2$ would give $0 \equiv 2$. Hence exactly $2 + \chi_2(q)$ classes, and the same count for C with -2 replaced by -4.

A cell of type D carries two linear conditions, on $M_i + 2$ and $M_i + 8 = M_{i+1} + 2$, and the partner $(M_i + 5)^2 - 11$. The linear classes $t \equiv -2, -8$ are distinct for $q > 3$, and $(t + 5)^2 \equiv 11$ contributes

$1+\chi_{11}(q)$ further classes, disjoint from them because either substitution gives $9 \equiv 11$, impossible for $q > 2$. Hence exactly $3+\chi_{11}(q)$. The same computation for E, whose offsets are $-2, -10$ and whose partner is $(M_i+6)^2 - 14$, gives $3+\chi_{14}(q)$; and for F, with offsets $-4, -8$ and partner $(M_i+6)^2-2$, gives $3+\chi_2(q)$. In every case the overlap check reduces to $16 \equiv 14$ or $4 \equiv 2$, impossible for $q > 2$. Write $\omega_X(q)$ for these counts.

Verification. Computed directly for all 423 primes $19 \le q < 3000$: zero disagreement with $\omega_A = \omega_C = 2+\chi_2$, $\omega_D = 3+\chi_{11}$, $\omega_E = 3+\chi_{14}$, $\omega_F = 3+\chi_2$.

Step 2: the sifting dimension is the mean of ω_X. I_X is contained in $\{0, 1, \ldots, N-1\}$ and avoids $\omega_X(q)$ classes modulo every prime $q > 17$; the primes $q \le 17$ contribute a bounded factor and are ignored. Since χ_2, χ_{11} and χ_{14} are real non-principal characters — of conductors 8, 44 and 56 — the series $\sum_q \chi_d(q)/q$ converges, by the non-vanishing of $L(1, \chi_d)$. Hence

$$\prod_{17<q<z}\Big(1-\frac{\omega_X(q)}{q}\Big) = \prod_{17<q<z}\Big(1-\frac{\kappa_X}{q}\Big)\cdot \prod_{17<q<z}\frac{1-\omega_X(q)/q}{1-\kappa_X/q} \asymp \frac{1}{(\log z)^{\kappa_X}},$$

with $\kappa_A = \kappa_C = 2$ and $\kappa_D = \kappa_E = \kappa_F = 3$, the second product converging because its logarithm is $-\sum_q \chi_d(q)/q + O(\sum_q q^{-2})$.

Step 3: the upper-bound sieve. Selberg's sieve applied to the interval $\{0, \ldots, N-1\}$ with the removed classes of Step 1 gives, for any z,

$$|I_X| \le N \prod_{q<z}\Big(1-\frac{\omega_X(q)}{q}\Big)(1+o(1)) + O(z^2).$$

Taking $z = N^{1/3}$ makes the error $O(N^{2/3})$ and, by Step 2, the main term $\ll N/(\log N)^{\kappa_X}$. This is the stated bound for each I_X.

Step 4: summation. Under the twinless hypothesis the type B is empty, since it requires both $M+2$ and $M+4$ prime. A sector carries at most one cell of any given type, so the number of exceptional cells in the block is exactly $\sum_X |I_X|$, which by Step 3 is $\ll N/(\log N)^2$. With $N = 35L$ this is $\ll L/\log^2 L$. ■

This is a computation, not a new tool, and it should be read as one. Reading the sieve dimension off the number of roots of the defining polynomials modulo each prime is the standard definition of dimension — the condition $\sum_{q<s}(\log q)/f(q) = \kappa \log s + O(1)$ of Halberstam and Richert — and the passage from a dimension-κ sifting density to an upper bound of order $N/(\log N)^\kappa$ is the standard Selberg estimate; see [V, ref. 8] for the higher-dimensional theory and [V, ref. 12] for the sieve itself. The twin problem in this language is the case $\rho(q) = 2$, dimension 2. What is particular to the present setting is only the input: that the exceptional cells of Theorem 4 have the five partners above, so that the quadratic characters χ_2, χ_{11} and χ_{14} appear and split the five types into dimensions $2, 2, 3, 3, 3$. The conclusion then follows by quoting the standard machinery, and we claim nothing for the machinery.

Two remarks on the shape of this. The bound is an **upper-bound sieve only**, and upper bounds in a fixed dimension are the direction in which sieve methods have no parity obstruction; nothing here bears on the lower-bound side. And the two-sector types are smaller than the one-sector types by a full logarithm, so in a long block almost every exception is of type A or C — a block that is longer is structurally *simpler*, not more complicated.

The first consequence is the one that matters here: **the density of exceptions tends to zero**, so the requirement per sector is not bounded below by any positive constant.

And the bound is numerically empty over every computable range. Taking $\rho^*(D) \sim D/\log D$ for the largest admissible set in an interval of diameter D, the crude form $B_L \le 2\rho^*(210L)-1$ gives, per sector, 2.22 at $L = 1$, 1.59 at $L = 9$, 1.06 at $L = 401$, 0.82 at $L = 10^4$ and 0.51 at $L = 10^8$: it does not fall below 1 until $L \approx 400$ and does not reach 1/2 until $L \approx 10^8$. The measured values are far below it everywhere in range, so the observed decrease is not this bound taking effect. Nor do the data identify the exponent: over $1 \le L \le 101$ the quantity $B_L \log^2(210L)/(210L)$ reads $4.22, 4.42, 4.61, 4.99, 4.91, 5.80, 6.68, 7.37$ — rising, not settling — while $B_L \log(210L)/(210L)$ reads $0.789, 0.685, 0.663, 0.685, 0.651, 0.691, 0.720, 0.740$. On this range $\log D$ varies only from 5.3 to 10.0, and the two laws are not separated by it.

What the whole calculation does not buy, stated exactly. Since $B_L = o(L)$, the criterion no longer asks for a survivor every sector or every second sector: it suffices to prove $\sum_i C_{M_i} \ge \varepsilon L$ for any fixed $\varepsilon > 0$, that is, that a positive proportion of sectors contains at least one open cell. Measured at $M_0 = 448{,}353$, the survivor total is 366,120, 1,100,106 and 3,304,035 over $L = 1, 3, 9$ — that is 10,461, 10,477 and 10,489 per sector, flat — so the inequality holds by a factor of 11,810, 16,419 and 20,270, widening with L. The survivor total is linear in L with a large constant; its logarithm is $\log M_0$, not $\log L$, so no amount of lengthening changes its shape.

> **The reduction is therefore real and it is not a route.** What is now required of the survivor side is not a rate of growth but an *existence* statement: one open cell in a positive proportion of sectors. By Theorem 4 an open cell is a twin pair unless it sits at one of six named places, and [V, §6.7] measures that discrepancy at 0, 1 and 3 over three full periods. So "prove $\sum_i C_{M_i} \ge \varepsilon L$" is "prove that a positive proportion of sectors contains a twin", and shrinking the right-hand side from 32 to $O(L/\log^2 L)$ changes the number on the right while leaving the missing ingredient on the left exactly as it was.

3.5 Theorem 5: a new line kills at most one pair in its own first window

The sector $[p^2, (p+2)^2)$ is where the line L_p is born. Index gap-2 pairs by their offset from the centre $C = (p+1)^2$, writing $P(j) = (C + 2j - 1,\ C + 2j + 1)$ with $-p \le j \le p$, so the window holds exactly $2p + 1$ pair slots. Let T_p^- count the surviving pairs before L_p is switched on, T_p^+ after, and $D_p = T_p^- - T_p^+$.

L_p has exactly three strikes in this window: p^2, $p(p+2)$ and $p(p+4)$, since $p(p+6) > (p+2)^2$.

Theorem 5. $D_p \in \{0, 1\}$ for every prime $p > 3$.

Proof. Three strikes, and each is disposed of by the grid L_3.

1. p^2**.** Of the two pairs it meets, only $(p^2, p^2 + 2)$ lies in the window — the other has index $-p - 1$. For $p > 3$, $p^2 \equiv 1 \pmod 3$, so $p^2 + 2 \equiv 0$: **that pair is already dead.**
2. **The strike divisible by** 3**.** One of $p+2, p+4$ is $\equiv 3 \pmod 6$, so one of $p(p+2), p(p+4)$ lies on L_3 and was struck before L_p existed; both pairs it meets were already dead.
3. **The surviving strike.** Call it $H_p = p(p+2)$ when $p \equiv 5 \pmod 6$ and $p(p+4)$ when $p \equiv 1 \pmod 6$. In either case $H_p \equiv 5 \pmod 6$, so $H_p - 2 \equiv 3 \pmod 6$ and the pair $(H_p - 2, H_p)$ is **also already dead.**

Only $(H_p, H_p + 2)$ remains. ■

Corollary 3. $D_p = 1$ if and only if **two** primality conditions hold together:

$$p \equiv 5\,(6): \quad p{+}2 \text{ prime and } p(p{+}2){+}2 \text{ prime}; \qquad p \equiv 1\,(6): \quad p{+}4 \text{ prime and } p(p{+}4){+}2 \text{ prime}.$$

Proof. H_p survives the older lines iff its cofactor ($p+2$ or $p+4$) has no prime factor below p; being less than $2p$, that means the cofactor is prime. And $H_p + 2 < (p+2)^2$ is not divisible by p, so if composite its least prime factor is below p and it was struck earlier; hence $H_p + 2$ survives iff it is prime. ■

Verification. Corollary 3 predicts D_p from two primality tests alone; checked against the directly computed D_p for every prime $5 \le p < 4000$, with **zero mismatches**. Examples: $p = 5$ gives 7 and 37 both prime, so $D_5 = 1$; $p = 11$ gives 13 prime but $145 = 5 \cdot 29$ composite, so $D_{11} = 0$; $p = 13$ gives 17 and 223 both prime, so $D_{13} = 1$.

What this does and does not buy. Since $T_p^+ = T_p^- - D_p$ and every survivor in the window is prime (the cofactor argument of §5.2), T_p^+ *is* the number of twin pairs in $[p^2, (p+2)^2)$. Theorem 5 therefore says:

$$\boxed{T_p^- \text{ is the twin count of the window, up to an error of 0 or 1, and the error is characterised.}}$$

That is the strongest local statement in this paper, and it is worth being explicit that it is not a reduction. T_p^- and T_p^+ differ by at most one, so proving anything about T_p^- is proving it about the twin count. In particular the sufficient condition "$T_p^- \ge 2$" is *stronger* than the conclusion "$T_p^+ \ge 1$", not weaker. What Corollary 3 adds is that even the discrepancy between the two is governed by a twin-like coincidence — two simultaneous primality conditions — so the error term is of the same nature as the quantity.

3.6 Theorem 6: the bridge pair, and the only law that knows about squares

Index gap-2 pairs by x as above but on an absolute scale, the pair being $(2x+1, 2x+3)$. The window for odd m begins at $a_m = (m^2-1)/2$ and holds $2m+1$ slots, while $a_{m+2} - a_m = 2m+2$. **So consecutive square windows do not abut: exactly one pair slot falls between them,**

$$g_m = a_m + 2m + 1, \qquad \text{the pair } \big((m+2)^2 - 2,\ (m+2)^2\big),$$

whose upper member is the next square itself. The line is partitioned as window, bridge, window, bridge, …, with no slack and no overlap.

Let B count the bridges surviving a line set, over a full cycle of roots.

Theorem 6. $B' = (q - 2 - \chi_q(2))B$, where $\chi_q(2)$ is the Legendre symbol: $+1$ for $q \equiv \pm 1 \pmod 8$ and -1 for $q \equiv \pm 3 \pmod 8$.

Proof. The bridge at root r dies under q when $q \mid r^2$, i.e. $r \equiv 0$ — one class — or when $r^2 \equiv 2 \pmod q$, which has $1 + \chi_q(2)$ solutions. The two conditions are disjoint for $q > 2$, so $2 + \chi_q(2)$ classes are lost. ■

Verification. $B = 2,\ 8,\ 32,\ 320,\ 3840,\ 53760,\ 967680$ for the line sets up to $3, 5, 7, 11, 13, 17, 19$, against $T = 1,\ 3,\ 15,\ 135,\ 1485,\ 22275,\ 378675$. Brute-forced from the definition for $\{3\}$, $\{3,5\}$, $\{3,5,7\}$, $\{3,5,7,11\}$: exact.

So the cycle fingerprint is not three numbers but four, with four distinct degrees:

$$(M, S, T, B) \longmapsto (qM,\ (q-1)S,\ (q-2)T,\ (q-2-\chi_q(2))B).$$

B **is the only one of the four that knows the window is anchored at a square**, and the arithmetic that enters is whether 2 is a quadratic residue.

Theorem 7 (the summation identity). Over M consecutive square windows,

$$\sum_{i=0}^{M-1} T_{m+2i} = 2(m+M)T - B.$$

Proof. The M windows together with their M bridges tile a stretch of $a_{m+2M} - a_m = 2M(m+M)$ pair slots, that is exactly $2(m+M)$ complete cycles, holding $2(m+M)T$ surviving pairs. The M bridge roots are M consecutive odd numbers, and since M is odd they cover every residue class modulo M exactly once, so exactly B of them survive. ■

Verification. Checked for $\{3,5\}$ and $\{3,5,7\}$ at $m = 9, 15, 101$ — six cases, all exact.

The collective bias, quantified. Against the density prediction $T(2m+2M-1)$ for the same total length, the windows hold exactly $T-B$ fewer pairs; measured sums of deviations $-1, -5, -17, -185$ for the four line sets, matching $T-B$ each time. Now

$$\frac{B}{T} = \prod_q \Big(1 - \frac{\chi_q(2)}{q-2}\Big),$$

which converges: measured $2.5554,\ 2.5517,\ 2.5614,\ 2.5615,\ 2.5622$ at $z = 19,\ 10^3,\ 10^4,\ 10^5,\ 10^6$. The reduced product $\prod(1-\chi_q(2)/q)$ reaches 1.604401 at $z = 10^6$ against

$$\frac{1}{L(1,\chi_8)} = 1.604556, \qquad L(1,\chi_8) = \frac{\log(1+\sqrt{2})}{\sqrt{2}} = 0.623225.$$

So the square geometry enters this framework through a Dirichlet L-value at 1 for the character modulo 8.

And the bias, though real, is not usable: $T - B \approx -1.56T$ is a fixed number independent of m, spread over M windows, so the deficit per window is $O(T/M)$ and vanishes against the per-window average as M grows.

3.7 The square-phase mean, and the end of the "poor window" question

One may ask directly whether windows anchored at squares are systematically poorer than windows placed anywhere. The answer is an identity rather than an experiment.

For a window starting at r^2, the pair at offset j survives q exactly when $r^2 \not\equiv -2j$ and $r^2 \not\equiv -2j-2 \pmod q$. Writing $\rho_q(a) = \mathrm{card}\{r : r^2 \equiv a\}$ and $\nu_q(j) = \rho_q(-2j) + \rho_q(-2j-2)$ — the two conditions cannot hold at once — the average over **all square phases** is exactly

$$\mu_\square(L) = \sum_{j<L} \prod_q \frac{q - \nu_q(j)}{q}. \qquad (3.3)$$

Computed against the naive density prediction $E = L\prod(q-2)/q$, and against the restricted average $\mu^\times$ over roots that are themselves coprime to every old line:

p	T_p^- (actual)	$\mu_\square$	$\mu^\times$	$E = L\delta$
11	2	2.943	2.833	3.286
29	2	4.149	4.201	4.206
53	2	5.434	5.283	5.457
101	7	7.653	7.473	7.774
499	13	21.237	21.319	21.274
997	38	34.591	34.534	34.608

$\mu_\square$ agrees with E to within about 1% from $p = 29$ onward, and restricting the roots changes nothing.

> **The square-phase mean has the exact expression (3.3); numerically it tracks the generic density prediction closely, the two differing by about 1% or less from $p = 29$ onward in the sample above.** So the exact quantity is available in closed form, and on the tested range it shows no bias at the level of the mean — which replaces the earlier control experiment with a computation, though not with a proof that the two agree in the limit. Individual windows are of course far from the mean — $T_{53}^- = 2$ against $\mu_\square = 5.43$ — but the scatter is governed by the correlation function of [II, Thm 4], not by any property of squares.

3.8 Raw cell count and main term

Bookkeeping used by the sections above.

$$C = \frac{v^2 - u^2}{6} - 1, \qquad C_- = 4n - 1(u = 6n - 1), \qquad C_+ = 8n + 3(u = 6n + 1),$$

and $C \equiv 3 \pmod 4$ always (zero failures over 427 consecutive prime pairs) — an exact property which nevertheless supplies no protective invariant, since a single line's deletion count has no fixed parity. The main term is $M = CP_2$.

3.9 Local deviations

Bookkeeping used by the sections above.

$$\varepsilon_r(u) = D_r(u) - \frac{2N_{r^-}(u)}{r}. \qquad (3.4)$$

Inside a sector, line r deletes at $s \equiv -u^2/6$ and $s \equiv (2 - u^2)/6 \pmod r$, and the gap between the two positions is exactly $3^{-1} \bmod r$ (zero failures among 85). **Every ruler's phase is therefore a function of the single quantity u^2**, and under $u \mapsto u + 6$ the phase moves *quadratically*, since $u^2 \mapsto u^2 + 12u + 36$. This quadratic motion is what makes cancellation possible at all.

4. The closing budget: τ against S

A change of object, stated before anything else. Sections 2 and 3 concerned twins: a surviving cell, whose two members differ by 2. The present section concerns a different graph — survivors joined to survivors at distance 6 — and the two must not be run together, because the word "gap 6" would otherwise cover both the *letter* 6 of §2.1 (a gap of 6 between consecutive odd composites, which contains a twin) and the *edge* of length 6 used here. **What an untouched edge exhibits once the remaining lines have acted is a prime pair $(p, p+6)$, not a twin pair.**

Measured, so that the distinction is not left rhetorical: inside $(P^2, 9P^2)$ at $P = 101$ there are 2,903 edges, of which 1,865 survive the remaining lines and 1,410 are genuine gap-6 configurations — for instance $(10247, 10253)$, $(10337, 10343)$, $(10601, 10607)$. **None of them is a twin.**

We keep the section because the budget it produces is exact, and because $(p, p+6)$ is open in precisely the same way and for precisely the same reason; but nothing here bears on the twin conjecture directly.

Instead of asking what the lines will close, we ask the dual question: **how many deletions are needed, at minimum, to close every distance-6 edge?** If the available deletions fall short, a pair survives.

4.1 Theorem 8 (the exact minimum cover)

Take survivors as vertices and join x to $x+6$. Let T, U, Q count the edges, the 3-term runs and the 4-term runs (runs, not components: a component on k vertices contributes $\max(0, k-2)$ to U and $\max(0, k-3)$ to Q).

> **Theorem 8.** The minimum number of deletions required to destroy every distance-6 edge is
> $$\tau = T - U + Q.$$

Proof. By Theorem 3 every component is a path on at most 4 vertices, and the minimum vertex cover of a path on k vertices is $\lfloor k/2 \rfloor$. Summing $(k-1) - (k-2) + (k-3)$ over components reproduces $\lfloor k/2 \rfloor$ for $k = 2, 3, 4$ — and **only** for those, since $k = 5$ would give 3 against the true value 2. The cap of Theorem 3 is thus exactly what makes the identity hold. ■

Verification by direct component decomposition:

lines	component census	min cover	$T - U + Q$
$\{5, 7\}$	$\{1{:}4, 2{:}4, 3{:}4, 4{:}6\}$	20	20
$\{5, 7, 11\}$	$\{1{:}68, 2{:}56, 3{:}44, 4{:}42\}$	184	184
$\{5, 7, 11, 13\}$	$\{1{:}1100, 2{:}788, 3{:}524, 4{:}378\}$	2068	2068

This is an exact combinatorial identity: no independence assumption and no density heuristic enters.

4.2 Propagation laws

Let $G = T - D$ count genuine gap-6 pairs, D those with a survivor between. *(The twins sit in D: the surviving middle differs by 2 from one of the two endpoints. So G — the object [V, §3.3] targets — is exactly the twin-free part, which is the content of the caution above. Incidentally, measured on all four cycles below, $U = D$ exactly; we do not use this.)*

> **Theorem 9.** On the full cycle, the entry of a new line r gives
> $$T' = (r-2)T, \qquad D' = (r-3)D, \qquad G' = (r-2)G + D.$$

Proof. Of the r copies of a pair, one has its left member struck and one its right, leaving $r-2$. A D-configuration has three sensitive positions (both ends and the middle), leaving $r-3$; and the copy whose middle is deleted becomes a genuine gap-6. ■

lines	V	T	D	G
$\{5\}$	8	6	4	2
$\{5,7\}$	48	30	16	14
$\{5,7,11\}$	480	270	128	142
$\{5,7,11,13\}$	5,760	2,970	1,280	1,690

Corollary 4. On the full cycle, $\dfrac{T}{V}=\rho_p=\prod_{5\le s\le p}\dfrac{s-2}{s-1}$ and $\dfrac{D}{T}=\theta_p=\dfrac{2}{3}\prod_{7\le s\le p}\dfrac{s-3}{s-2}$ **— exact identities, not estimates.**

p	23	53	101	199	499	997
ρ_p	0.4358	0.3607	0.3151	0.2746	0.2367	0.2138
θ_p	0.3606	0.2967	0.2588	0.2253	0.1941	0.1753

Both tend to zero: distance-6 pairs become rarer, yet a growing share of those remaining become genuine gaps.

4.3 Theorem 10 (tail compression)

Inside $(P^2, 9P^2)$, after the lines up to P have acted, every surviving composite has the form $n=qr$ with

$$P<q<3P, \qquad q\le r<\frac{9P^2}{q}<9P.$$

Indeed three factors above P would give $n>P^3>9P^2$ for $P>9$, and the smaller of the two remaining factors is below $3P$. Consequently the number of genuinely new strikes contributed later by a line q with $P<q<3P$ is

$$E_P(q)=\pi(9P^2/q)-\pi(q-1)<2(3P-q)+1,$$

so a line approaching $3P$ loses power because the available cofactor interval contracts.

A related compression occurs among the **late lines in the sweep up to P itself**:

> **Theorem 10.** For $P\ge 243$ and $P/3<q<P$, every new strike of q inside the window has the form $x=qr$ with r **prime**.

Proof. The cofactor satisfies $r<9P^2/q<27P$. If r were composite, all its prime factors would be $\ge q$ (it survived the smaller lines), so $r\ge q^2>P^2/9$. The contradiction holds precisely when $P^2/9\ge 27P$, i.e. $P\ge 243$. ■

Finite check around the threshold. Direct enumeration of new strikes with composite cofactor:

P	101	151	199	211	**241**	251	307	499	997
anomalies	26	9	6	5	**0**	0	0	0	0

The enumeration exhibits anomalies at smaller P and none in the tested cases from 241 onward; this is consistent with, but stronger numerically than, the proved sufficient threshold 243. The sharper pointwise condition is $q^3 > 9P^2$, i.e. $q > 9^{1/3}P^{2/3}$; and the two conditions cross exactly at

$$\frac{P}{3} = 9^{1/3}P^{2/3} \iff \frac{P^3}{27} = 9P^2 \iff P = 243,$$

so 243 is the point at which the constraint $q > P/3$ becomes the binding one.

5. Clocks, and primality as a zero-test

5.1 Theorem 11 (the shift law) and primality

Theorem 11. $\phi_q(p+2) = \phi_q(p) - 1 \pmod{q}$ for every old line q; and when p itself becomes an old line, $\phi_p(p+2) = p - 1$.

Proof. Substituting $p+2$ for p in [III, (2.2)] subtracts 1. For the second, $\phi_p(p+2) = (p-(p+2))/2 = -1 \equiv p - 1$. ■

Verification. Zero failures among 4,983 instances, and zero for the insertion rule.

Hence, writing $\Phi(n) = (\phi_3, \phi_5, \phi_7, \dots)$, the transition $n \mapsto n+2$ acts as

$$\Phi \longmapsto \Phi - 1, \qquad (5.1)$$

each component on its own circle, and the whole system evolves by three small numbers:

step $+4$, first square gap $+8$, all clocks -1,

with one clock inserted at $p-1$ whenever p is prime. The system is *shift, zero-test, insert*; nothing is rebuilt.

Corollary 5. p is composite if and only if some old clock reads 0 at its birth. Equivalently, **a prime is a step at which no clock lands on zero.**

Verification. Zero failures over all odd $p < 2000$.

Corollary 6. The clock of L_3 cycles $2 \to 1 \to 0$, so the two non-zero states are exactly the cell $(6a-1, 6a+1)$. **The cell, taken as a definition in [I, §3], is a consequence.**

5.2 Theorem 12 (surviving cofactors are prime)

Theorem 12. For $p \geq 11$ and $p < t \leq 9p$, a cofactor t surviving all lines below p is prime.

Proof. $t \leq 9p < p^2$. A composite t surviving all lines $< p$ would need two prime factors $\geq p$, giving $t \geq p^2 > 9p$. ■

Verification. Zero violations at $p = 11, 13, 17, 101, 499$.

Hence J_j, the number of new strikes in sector j, equals the number of **primes** among that block's cofactors. Primality appears as *a slot no ruler reached*, with no change to its definition.

Worked example ($p = 11$, computed without writing a single strike).

quantity	value
κ_j	0, 2, 4, 7, 10, 14, 18, 22, 27, 32, 38, 44
H_j	2, 2, 3, 3, 4, 4, 4, 5, 5, 6, 6
clocks	$\phi_3 = 2$, $\phi_5 = 2$, $\phi_7 = 5$
J_j	1, 2, 1, 2, 1, 3, 1, 2, 3, 2, 2

The "new" cofactors are exactly $13, 17, 19, 23, \dots, 97$, all prime, as Theorem 12 requires. ### 5.3 The general form: a row reads primality off originality

Theorem 12 is the case that arises inside one sector. Read along the whole row of p it has a general form, and the general form is worth stating because it makes the twin condition geometric.

Call a cell of the row **original** if no line below p owns it. Everything before p^2 is inherited — a strike pm with $m < p$ carries the smallest prime factor of m, which is below p — so p^2 is the first original cell on the row. Past it:

Theorem 13. For p prime and $p \le p + 2j < p^2$,

$$p(p+2j) \text{ is original with respect to the lines below } p \quad \Longleftrightarrow \quad p+2j \text{ is prime.}$$

Proof. If $p + 2j$ is composite and below p^2 its least prime factor is below p, so that line already owns the cell. If $p + 2j$ is prime it has no factor below p at all, and p itself begins at p^2. ■

Verification. Zero failures over 1,782,933 instances: every prime $p < 400$ and every j with $p+2j < p^2$.

So a row is a reader. The row of p carries, in the originality of its cells, the primality of every odd number from p up to p^2 — one bit per cell, with no change to the definition of a prime. Two consequences are worth recording.

First, the twin condition becomes a two-step statement at the diagonal. The cells at p^2 and $p(p + 2)$ are the first two on the row past the inherited region, and by Theorem 13

$$(p, p+2) \text{ is a twin pair} \quad \Longleftrightarrow \quad \text{originality survives one step past } p^2.$$

Writing O for original and I for inherited, the row crosses the diagonal as . . . I | OO . . . at a twin and . . . I | OI . . . otherwise: at $p = 23$, 529 is original but $575 = 23 \cdot 5^2$ is not, and $(23, 25)$ is not a twin.

Second, the shadow lies on a curve already in the series. The tested cell is

$$p(p+2) = (p+1)^2 - 1,$$

so every twin test sits one unit below an even square: $35 = 36 - 1$, $143 = 144 - 1$, $323 = 324 - 1$, $899 = 900 - 1$. **That is [I, Thm 5] read along the row instead of across the window** — the cell $(n^2 - 1,\ n^2 + 1)$ with $n = p+1$, whose lower member $(n-1)(n+1)$ is composite by construction. The two statements are the same fact.

And the reformulation is exact, which is precisely why it is not progress. Writing $\sigma(p)$ for the smallest line owning $p(p + 2)$, one has $\sigma(p) = \operatorname{spf}(p + 2) \le \sqrt{p+2}$ when $p + 2$ is composite, and $\sigma(p) = \infty$ exactly when $(p, p + 2)$ is a twin. Proving $\sigma(p) = \infty$ infinitely often is proving the twin conjecture, in the same words. What the row picture adds is a suggestion — that if twins were finite, every new diagonal point would need its shadow claimed by some line below $\sqrt{p}$ — and the suggestion does not survive measurement: over 6,835 composite cases with $p < 200{,}000$ the claimant is at most 13 in 63.6% of them and at most 100 in 89.9%, the counts being 5: 2248, 7: 1125, 11: 557, 13: 415, 17: 303. **The $\sqrt{p}$ bound is nowhere near tight; the shadows are claimed by the smallest lines, not by a delicate conspiracy of many.**

6. Inheritance across sectors, and the capacity of a single line

Sections 2–5 treat one sector at a time. This section treats the *sequence* of sectors, indexing them by $M = 9, 15, 21, \ldots$ — the odd multiples of 3 — with the sector $(M^2, (M+6)^2)$ carrying $A(M) = 2M + 6$ cells. Three exact laws come out, all sharper than the average statement "line q removes $2/q$ of what remains", because each is a statement about a *named* gap rather than about a count.

6.1 Theorem 14: the sector inheritance law

Fix any finite set of lines $p_1 < \ldots < p_r$ above 3, and put

$$Q = \prod_i p_i, \qquad S = \prod_i (p_i - 2),$$

so that S cells survive that set in each cycle of Q consecutive cells. Let $B(M)$ be the number of cells of the sector at M that survive those lines.

Theorem 14. $B(M + 6Q) = B(M) + 12S$.

Proof. Two facts. First, the sector grows by exactly $12Q$ cells: $A(M + 6Q) - A(M) = 12Q$. Second — and this is what makes the law exact rather than approximate — the **phase is preserved**: the sector at M begins near cell $M^2/6$, and

$$\frac{(M+6Q)^2}{6} - \frac{M^2}{6} = 2MQ + 6Q^2 \equiv 0 \pmod{Q}.$$

So the first $A(M)$ cells of the later sector repeat the earlier sector's pattern exactly, and the tail of $12Q$ new cells is precisely twelve complete cycles, each leaving S survivors. ■

Verification. Exact at every M tested, for each of the three sets: $\{5\}$ ($Q = 5$, $S = 3$, increment 36); $\{5, 7\}$ ($Q = 35$, $S = 15$, increment 180); $\{5, 7, 11\}$ ($Q = 385$, $S = 135$, increment 1620).

What it says. A *fixed* set of old lines never catches up with the window. Each time its cycle returns to the same phase, the sector has grown, and a known positive number $12S$ of fresh open cells appears. Only lines born after the set was fixed can close them.

6.2 Theorem 15: the capacity of one new line on one family

Each survivor of the fixed set appears in the new tail exactly twelve times, at cells

$$x,\ x + Q,\ x + 2Q,\ \ldots,\ x + 11Q,$$

which we call a **family**. A new line q closes a cell c when $c \equiv \pm 6^{-1} \pmod{q}$, so on a family it closes the copies t solving $x + tQ \equiv \pm 6^{-1}$. There are two such t modulo q, and their separation does not depend on x:

Theorem 15. Put $\Delta_q \equiv (3Q)^{-1} \pmod{q}$ and $d_q = \min(\Delta_q,\ q - \Delta_q)$. Then a line $q > 11$ closes at most two copies of any family, and **at most one** whenever $d_q > 11$.

Proof. The two solutions differ by $2 \cdot 6^{-1}Q^{-1} = (3Q)^{-1}$, whose least absolute representative is $\pm d_q$. Two copies lie in the family only if two values of $t \in \{0, \dots, 11\}$ differ by d_q, which needs $d_q \le 11$. ■

Verification. For $Q = 385$, brute force over all x and all q from 13 to 101 reproduces the predicted capacity with **zero mismatches**. Sample values of d_q: 13:6, 17:1, 19:5, 23:9, 31:4, 37:14, 53:24, 83:12, 101:39.

6.3 Corollary 7: the exceptional lines are finite in number

Corollary 7. A line q can close two copies of a family only if $q \mid 3Qr \pm 1$ for some $1 \le r \le 11$. Consequently every $q > 33Q + 1$ closes **at most one** copy of every family.

Proof. $d_q \le 11$ means $\Delta_q \equiv \pm r$ with $r \le 11$, i.e. $3Qr \equiv \pm 1 \pmod q$; and $0 < 3Qr \mp 1 \le 33Q + 1$, so q cannot divide it once q exceeds that bound. ■

For $Q = 385$ the threshold is 12,706. This is a genuinely local statement: it names, for each family, a bound on what a *specific* line can do, whereas $2/q$ only bounds a total.

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7. The gate belt: what a line can do between its own square and the next

Sections 3–6 work inside a sector bounded by consecutive odd squares. This section changes the unit: since every prime $q > 3$ has $q^2 \equiv 1 \pmod 6$, each prime has a **gate** G_q with $q^2 = 6G_q + 1$, and the cell $C_{G_q} = (q^2 - 2, q^2)$ is closed by q itself. Consecutive primes $q < r$ therefore delimit a **belt** of cells $C_{G_q+1}, \dots, C_{G_r-1}$ between two gates that are certainly closed. The belts tile the cell axis.

7.1 Theorem 16: the size of a belt

Theorem 16. For consecutive primes $q < r$ with $g = r - q$, the belt holds exactly

$$G(q,r) = \frac{r^2 - q^2}{6} - 1 = \frac{g(2q+g)}{6} - 1$$

cells, so its size grows like qg.

Proof. Both q^2 and r^2 are $\equiv 1 \pmod 6$, so the cells strictly between the two gates are exactly the $(r^2 - q^2)/6 - 1$ complete cells of L_3 in the interval. ■

Verification. $G = 3, 11, 7, 19, 11, 27, 51, 19, 67$ for the nine consecutive belts $5 \to 7, 7 \to 11, \dots, 31 \to 37$, and $247, 67, 667, 4011$ for $89 \to 97$, $101 \to 103$, $499 \to 503$ and $997 \to 1009$; each reproduces from the formula.

7.2 Verified Law 17: the new line's reach depends on the gap, not on its size

Verified Law 17 (conditional). Suppose $g^2 < 2q$. Then, within its own belt, the number of cells the line L_q can strike at all is

$$H_q = \left\lfloor \frac{2g}{3} \right\rfloor,$$

independent of q.

Proof under the hypothesis. If $g^2 < 2q$ the line completes no extra lap before the next gate, so it reaches only the cofactors $q+2, q+4, \ldots, q+2g$, giving g candidate strikes; one in every three falls on L_3 and so touches no cell of the grid, leaving $\lfloor 2g/3 \rfloor$. ■

We do not call this a theorem, because the hypothesis is not available. $g^2 < 2q$ is far weaker than Cramér's conjecture $g = O(\log^2 q)$, which would give it at once — but it is **stronger than anything currently proved, and stronger than the Riemann hypothesis supplies**: RH gives only $g \ll \sqrt{q}\log q$, hence $g^2 \ll q\log^2 q$, which does not suffice. **Verified over 17,981 belts, every consecutive prime pair with $q <$ 200,000: no failure.**

The consequence is worth stating plainly. The belt has $\sim qg/6$ cells and the line born at its left end can touch $\sim 2g/3$ of them. **For a twin gap $g = 2$ the line touches exactly one cell, however large q is** — one cell out of $\sim q/3$. A line at $q \approx 10^6$ entering a belt of some hundred thousand cells has a single strike available before the next gate opens.

7.3 The collapse of the new line's effect

Raw reach is not closing power: a strike may land on a cell an older line has already closed. Write K_q for the cells the new line closes **first**.

belt	G	H_q	K_q	twins left
$5 \to 7$	3	1	1	2
$7 \to 11$	11	2	2	4
$11 \to 13$	7	1	**0**	2
$13 \to 17$	19	2	1	7
$17 \to 19$	11	1	**0**	2
$23 \to 29$	51	4	**0**	8
$31 \to 37$	67	4	**0**	11
$89 \to 97$	247	5	**0**	21
$101 \to 103$	67	1	**0**	7

Measured over every consecutive prime pair below 5,000*:* $K_q = 0$ in 71.1% of belts with $q < 1000$ and 76.8% of belts with $1000 < q < 5000$; mean K_q falls from 0.331 to 0.259; the maximum ever observed is 3. **A new line typically arrives at its own gate to find that the work has already been done.**

A monotone version of this is false, and we record it because it is the natural guess. It is not the case that a newer line always closes fewer cells than every older one: in the belt $31 \to 37$ the first closures are $17:1$, $19:3$, $23:2$, $29:3$, so 29 — newer than 19 and 23 — closes more than both. **The weakness is collective, not line by line.**

7.4 A deterministic ceiling for a whole age layer

Nothing above uses primality of the intermediate lines, and the next bound deliberately gives them more power than they have.

> **Proposition 2.** In the belt $q \to r$ of length $L = r^2 - q^2$, any line s makes at most $\lceil L/2s \rceil$ strikes, of which at most a fraction $2/3$ touch cells of the grid. Hence, allowing

every odd s not divisible by 3 in a range to act as an independent line and ignoring all overlap between them, the layer $q-D \le s \le q$ can close at most

$$C_D(q,r) = \sum_{\substack{q-D\le s\le q\\ s\text{ odd},\ 3\nmid s}} \left\lceil \tfrac{2}{3} \left\lceil \tfrac{L}{2s} \right\rceil \right\rceil$$

cells.

For the belt $499 \to 503$ ($G = 667$): the newest quarter has ceiling $C = 168$ (25% of the belt) and closes 6 in fact; the newest half has ceiling 376 (56%) and closes 15. For $997 \to 1009$ ($G = 4011$): the newest tenth has ceiling 316 and closes 14; the newest quarter 836 and closes 28; the newest half 1,977 — under half the belt — and closes 68. **The ceilings are generous by one to two orders of magnitude, and the real burden falls on lines far below $q/2$.**

7.5 And why the layer ceilings do not close the argument

Proposition 2 invites an obvious attempt: build the full pyramid of age layers $(q/2, q]$, $(q/4, q/2]$, ... down to $s = 5$, sum the ceilings, and hope the total falls short of G. **It does not.**

belt	G	$\sum$ ceilings over all layers	ratio
$499 \to 503$	667	2,094	3.1×
$997 \to 1009$	4,011	13,935	3.5×
$10007 \to 10009$	6,671	35,045	5.3×

and the cumulative total already passes G at the **second** layer.

The asymptotic is worth getting right, because it is the point of the section. **The sum runs over every s coprime to 6, not over the primes**, and those have density $1/3$, so

$$\sum_{\substack{s\le q\\ (s,6)=1}} \frac{1}{s} = \frac{1}{3}\log q + O(1), \qquad \text{whence} \qquad \sum_s \tfrac{2}{3}\cdot\tfrac{L}{2s} = \frac{L}{3}\sum_s \frac{1}{s} \sim \frac{L}{9}\log q.$$

Against $G \sim L/6$ the ratio is therefore

$$\frac{\sum_s C_s}{G} \sim \frac{2}{3}\log q$$

— a **logarithm**, not an iterated logarithm. Checked against the table: $\frac{2}{3}\log(q/4)$ gives 3.22, 3.68, 5.22 at $q = 499,\ 997,\ 10007$ against the measured 3.14, 3.47, 5.25.

> **So the belt decomposition establishes three of the four things one would want — the belt grows like qg, the new line's reach is $O(g)$ and independent of q, and no fixed set of old lines can serve arbitrarily long belts — and refutes the fourth. The moving tail of recent lines is not capacity-limited; its ceilings exceed the belt by a factor that grows.** What is left is the forced overlap between the layers, which is $\prod(1-2/q)$ and describes the cycle, not the belt. This is [V, §6] again, reached from the belt side.

8. Four named cells inside the window

Paper I, §4, indexes the window by its own cell numbers: it is the interval $c_0, \dots, c_0 + N - 1$ with $c_0 = 6a^2 - 2a + 1$ and $N = 4a - 1$, where $n = 6a$ and the window is $[(n-1)^2, (n+1)^2]$. This section uses that indexing to study the four cells nearest its two ends.

A note on notation. The four cells of this section are written $T_1, \dots, T_4$ and are **not** the exception types A, C, D, E, F of §3.3 and §3.4; the two families are unrelated, and the letters are kept apart on purpose.

8.1 The four tracks, their character conditions and their densities

The window's template [I, §4.2] singles out four cells near its two ends. Writing $q = 6a - 1$ they are

$$T_1 = (q^2{+}4,\ q^2{+}6), \quad T_2 = (q^2{+}10,\ q^2{+}12), \quad T_3 = ((q{+}2)^2{-}14,\ (q{+}2)^2{-}12), \quad T_4 = ((q{+}2)^2{-}8,\ (q{+}2)^2{-}6),$$

and as a runs they trace four **tracks**. Substituting $q = 6a - 1$ makes every member a quadratic in a:

cell	lower member	upper member
T_1	$36a^2 - 12a + 5$	$36a^2 - 12a + 7$
T_2	$36a^2 - 12a + 11$	$36a^2 - 12a + 13$
T_3	$36a^2 + 12a - 13$	$36a^2 + 12a - 11$
T_4	$36a^2 + 12a - 7$	$36a^2 + 12a - 5$

Verification. Exact for $a = 1, \dots, 399$.

Theorem 18 (which lines can ever own a track). A prime r divides $36a^2 + Ba + C$ for some a exactly when the discriminant $B^2 - 144C$ is a quadratic residue modulo r. For the eight members the discriminants are $144k$ with

$$k = -4,\ -6\ (T_1); \qquad -10,\ -12\ (T_2); \qquad 14,\ 12\ (T_3); \qquad 8,\ 6\ (T_4),$$

so the conditions read $r \equiv 1 \pmod 4$ and $(-6|r) = 1$ for T_1; $(-10|r) = 1$ and $r \equiv 1 \pmod 3$ for T_2; $(14|r) = 1$ and $(3|r) = 1$ for T_3; $r \equiv \pm 1 \pmod 8$ and $(6|r) = 1$ for T_4.

Verification. Every prime factor of every member for $a = 1, \dots, 400$ — 4,209 checks — satisfies its condition; no violation.

Each individual condition admits half the primes (measured over primes below 10^5: 49.7–50.0%), but a cell falls to a strike on **either** member, so the union admits three quarters: measured 75.0, 74.8, 75.0, 75.0% for T_1, T_2, T_3, T_4. Requiring eligibility for all four at once cuts this to **exactly a quarter** — the eight discriminants reduce to the five independent characters $(-1), (2), (3), (5), (7)$, giving 32 sign patterns of which 8 pass; measured 24.84% against the naive independent guess $(3/4)^4 = 31.6\%$.

The four tracks are not equivalent. Each is a pair of quadratics, so its twin density is governed by a Bateman–Horn constant $S = \prod_r (1 - \nu_r/r)/(1 - 1/r)^2$, where ν_r counts the roots of the pair modulo r. The correct baseline is a generic cell $(6c-1, 6c+1)$, whose constant is $12C_2 = 7.9219$ — **not** the twin constant $2C_2 = 1.320$, which is for pairs $(n, n+2)$ over all n and counts the even n a cell never has.

track	ν_5	ν_7	ν_{11}	S	$S/12C_2$
T_1	**4**	2	2	3.230	0.408
T_2	1	4	2	5.797	0.732
T_3	2	1	4	8.739	1.103
T_4	2	2	**0**	11.324	1.429

Verification. Predicted density $S/\log^2(36a^2)$ against measured, for $a = 12{,}000, \ldots, 30{,}000$: 0.0059/0.0062, 0.0105/0.0100, 0.0158/0.0156, 0.0205/0.0214.

So track T_4 is 3.5 times richer in twins than track T_1, and the reason is visible in the table: $\nu_{11} = 0$ for T_4 — eleven never divides either of its members — while $\nu_5 = 4$ for T_1, the maximum, five dividing both members with two roots each. *(This is directly usable: a search for twin pairs near squares is three and a half times more productive on the T_4 track than on the T_1 track.)*

8.2 Theorem 19: simultaneity, and why it is the sharp question

Eligibility asks which primes can own a track at **some** a. The sharper question is which can own two tracks at the **same** a, and the answer is finite.

Theorem 19. A prime $r > 3$ can close two of T_1, T_2, T_3, T_4 in the same window only if it divides the resultant of the corresponding pair of quadratics. The complete list is

pair	admissible primes
T_1 & T_2	**none**
T_1 & T_3	5, 11, 13, 73
T_1 & T_4	5, 7
T_2 & T_3	5, 7, 11, 13, 37
T_2 & T_4	7, 19, 89, 97
T_3 & T_4	**none**

so the union is the nine primes $\{5, 7, 11, 13, 19, 37, 73, 89, 97\}$, and **every $r > 97$ closes at most one of the four cells in any single window.**

Proof of the two empty entries. The differences between a member of T_1 and a member of T_2 are 4, 6 and 8; a prime dividing one member of each would divide one of these, impossible for $r > 3$. The same three differences occur between T_3 and T_4. ∎

Proof of the rest. Two quadratics with the same leading coefficient differ by a linear form, so a common root modulo r forces a linear congruence in a; substituting it back leaves a fixed integer that r must divide. For T_1 lower against T_3 lower, for instance, $24a \equiv 18$ gives $4a \equiv 3$ and then $r \mid 65$. Each entry above was computed as the resultant and then checked for a genuine common root. ∎

The contrast is the point. Eligibility for one cell admits three quarters of all primes; for all four at once, a quarter — both infinite. **Simultaneous double duty admits nine primes and no more.** The character condition loses the shared variable a; restoring it collapses an infinite set to a finite one, and this is the sharpest local statement in the paper after Corollary 7.

And, as with every local statement here, it does not bind. Closing all four cells requires at least two lines — a special prime may serve T_1 & T_3 and another T_2 & T_4 — and at most four. Measured over $a = 3000, \ldots, 10000$: of 7,000 windows, 6,512 have all four closed, using two distinct lines in 973 cases, three in 4,748 and four in 791, so one of the nine special primes does double duty in 5,721 of them. Against this, the lines available number $\pi(q) = 428$, 2,062 and 6,055 at $a = 500$, 3000, 10000. **Four out of six thousand is free.**

8.3 Proposition 3: and why Theorem 19 does not obstruct anything

Theorem 19 is sharp, and it is sharp for one line. The next statement shows that it dissolves the moment one is allowed four, and it dissolves by construction rather than by measurement.

> **Proposition 3.** Let k tracks be given, each a pair of quadratics in a, and let N be any bound. Then there is an arithmetic progression of a — infinite, explicit, and computable — along which all k tracks are closed simultaneously, every closing line exceeding N and all k of them distinct. Any finite number of further congruence conditions may be imposed at the same time.

Proof. For each track choose a prime $r_i > N$, distinct from the others, whose discriminant condition (Theorem 18) is satisfied, and a root a_i of one of its members modulo r_i. The k conditions $a \equiv a_i \pmod{r_i}$ have pairwise coprime moduli, so the Chinese remainder theorem combines them into a single class modulo $\prod r_i$. Further conditions on coprime moduli are appended the same way. ■

The contrast with Theorem 19 is the whole point. There the same line had to satisfy two conditions *at the same* a, which is a genuine constraint and collapsed an infinite set to nine primes. Here the conditions sit on different moduli, and the shared variable costs nothing.

Explicit instance, with every step verified. Take

$$101 \mid T_1^-,\ a \equiv 54; \qquad 103 \mid T_2^-,\ a \equiv 15; \qquad 107 \mid T_3^-,\ a \equiv 73; \qquad 113 \mid T_4^-,\ a \equiv 77,$$

four distinct lines, all above 97. The Chinese remainder theorem gives

$$a \equiv 107{,}106{,}110 \pmod{125{,}782{,}673},$$

and along this progression T_1, T_2, T_3 and T_4 are all closed. Adjoining the further condition $5 \mid q+2$, i.e. $a \equiv 4 \pmod 5$ — which makes the new line's own central strike $q(q+2)$ inherited rather than new, so that the centre is not a twin either — gives

$$a \equiv 484{,}454{,}129 \pmod{628{,}913{,}365}, \qquad \text{i.e.} \qquad q \equiv 2{,}906{,}724{,}773 \pmod{3{,}773{,}480{,}190}.$$

The residue and the modulus are coprime, so by Dirichlet's theorem the progression contains infinitely many primes q. **Along it, q is prime, $q+2$ is composite, and all four named cells are closed — permanently and by construction.**

> **So no fixed number of named cells can force a twin.** Whatever finite list of tracks one selects, one distinct line may be assigned to each and the conditions combined; the construction is immune to how large the tracks' moduli are required to be, and it survives the addition of any finite list of side conditions. **An argument of this shape can only begin to bite when the number of cells grows with the window**, so that the number of conditions grows too and the assignment of a private line to each ceases to be free.

We state this as a proposition rather than a remark because it is the reason to stop, and knowing why one stops is worth more than another negative measurement.

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9. What this paper establishes, and what it does not

Proved here. The gap alphabet and the ladder (Theorems 1–3, Corollary 1); the dichotomy that three open exception positions occur only when $(M+2, M+4)$ is itself a twin (Corollary 2); the exception budget of 31 over a full period, and the proof that no finite set of lines lowers it (§3.3); the block budgets $B_3 = 67$ and $B_5 = 100$, and the sieve dimensions of the five exception types together with the resulting order $B_L \ll L/\log^2 L$ (Proposition 1, §3.4); the belt size and the layer ceiling (Theorem 16, Proposition 2); the character conditions on the four outer tracks, the nine-element simultaneity set, and the construction that shows it obstructs nothing (Theorems 18, 19, Proposition 3); the six exception positions (Theorem 4); the single-kill bound in a line's own first window and its characterisation (Theorem 5, Corollary 3); the bridge law and the summation identity over a cycle of windows (Theorems 6, 7); the exact minimum cover and the propagation and compression laws (Theorems 8–10, Corollaries 3–5); the shift law, the primality of surviving cofactors, and the originality law on a row (Theorems 11, 12, 13); and the sector inheritance and single-line capacity laws (Theorems 14, 15, Corollary 7).

Stated but not yet audited by the verification script. `verify_exception_dichotomy.py` covers the closed forms of the six positions, the $\{5, 7\}$ table, Corollary 2, and Step 1 of Proposition 1 (the forbidden-class counts, over all 423 primes below 3000). The block values $B_7 = 138$ and $B_9 = 163$ of §3.4 were computed with pruning and are stable across twelve and eight consecutive primes respectively; they should be read as lower bounds with strong stability, not as certified maxima, and the same applies to the values quoted there for $L = 21, 51, 101$. The sector-coupling budget of §3.3, the admissibility of the surviving configuration and the emptiness of the timing table are exhaustive finite computations, and each is a proof, but they are not yet in the script; until they are, they should be read as stated rather than audited.

Verified but not proved. Verified Law 17 (the reach $\lfloor 2g/3 \rfloor$), proved under $g^2 < 2q$ — a hypothesis weaker than Cramér but stronger than the Riemann hypothesis supplies; checked on 17,981 belts with $q < 200{,}000$ without failure.

Not here. Every one of these is an *exact* statement — an identity, a cap, or an explicit list. None is a lower bound, and a lower bound is what the twin conjecture needs. Each of the criteria above turns out, on inspection, to require a lower bound on a quantity that exceeds the twin count of a sector by at most a constant; the criteria are therefore exact reformulations rather than routes.

The measurements that establish that, the two test cases against which the framework was checked, and the account of where and why it stops, are **Paper V**.

References

The companion papers are cited as [0], [I], [II], [III], [V]. This paper imports only their definitions and proves everything else; the external literature is discussed in Paper V, where the framework is compared with it.